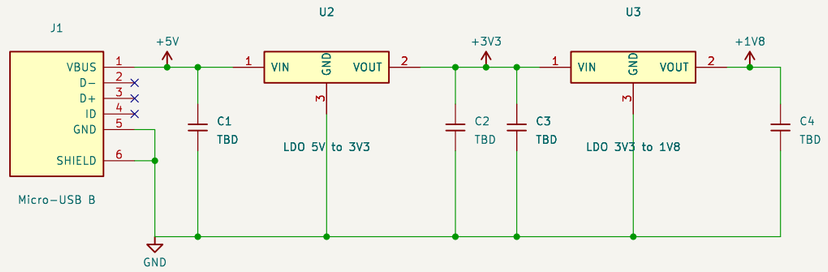
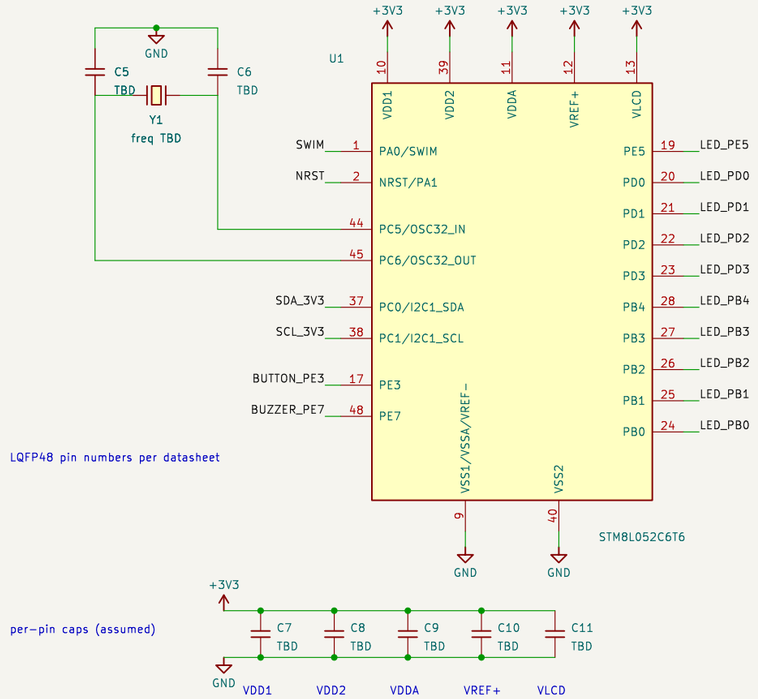
1 Power input and regulators



U2 and U3 are SOT−23 LDOs; U3 is fed from the 3V3 rail and powers the SGP30. GND is shared.

USB: VBUS and GND connected, shield tied to GND; D−, D+, ID floating

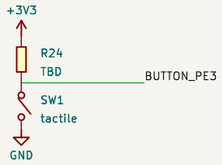
2 MCU, crystal and decoupling



LQFP48 pin numbers per datasheet

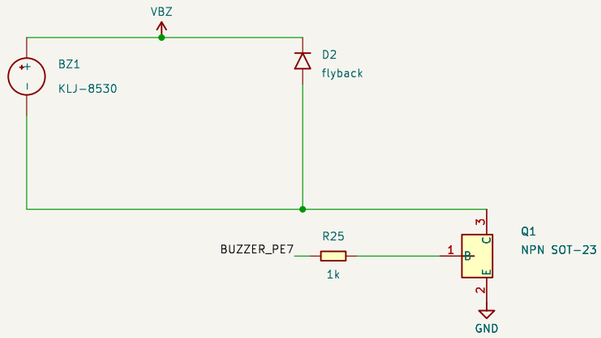
per−pin caps (assumed)

3 Button



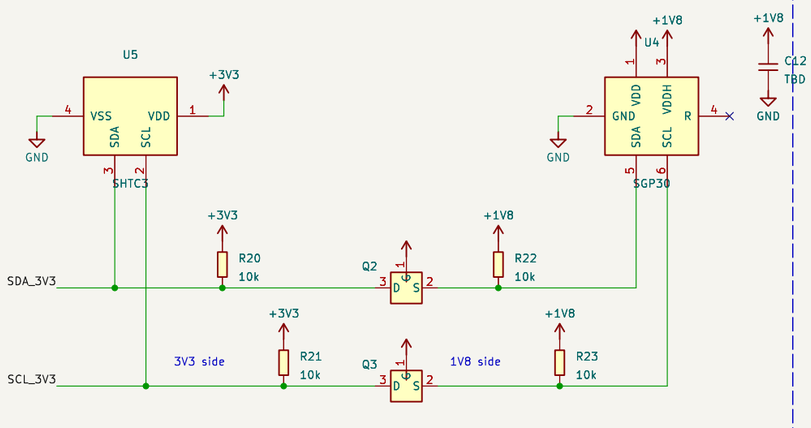
Switch on PE3 (back side). R24 pull−up assumed (0402 next to the switch)

4 Buzzer driver



Buzzer on PE7 (net BUZZER_PE7), R25 = 1k measured. Supply rail VBZ and the driver details are assumed

5 I2C bus, level shifter and sensors

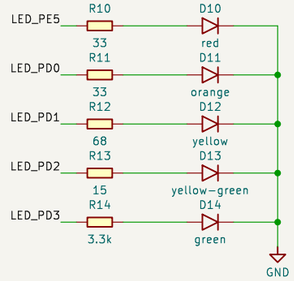


Q2/Q3: the two SOT−23 beside the SGP30, assumed to be the I2C level shifter (to confirm). Pull−ups 10k (measured).

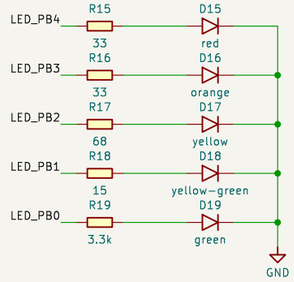
SGP30: VDDH tied to 1V8 (assumed), R pin open, pin numbers to verify. SHTC3 on 3V3 (assumed), on an isolated finger.

6 LED bars (10 LEDs on GPIO)

CO2 bar (top to bottom)



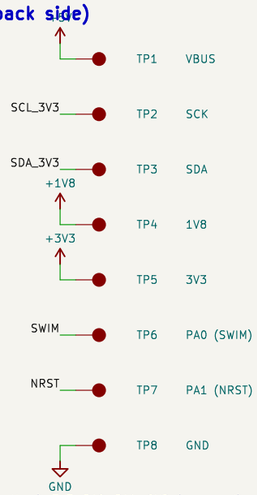
VOC bar (top to bottom)



Series R in ohm (measured); anode side GPIO, cathode GND.

Green LEDs use 3.3k, the others 15 to 68 ohm.

7 Test pads (back side)



Vertical group (top to bottom): 3V3, PA0, PA1, GND (measured)

Notes

Draft 0.4. Reconstructed from photos, manual measurements and the STM8L052C6 datasheet. Not verified against the real board.

Measured: both LDOs (5V to 3V3 to 1V8), I2C pull−ups 10k, buzzer on PE7 with R25 1k, switch on PE3, LED ports, colors and series resistors, test pad to port connections, USB pins and shield.

Measured: all five MCU supply pins (VDD1, VDD2, VDDA, VREF+, VLCD) at 3V3; five decoupling capacitors near the MCU. Datasheet: PA0 is SWIM, PA1 is NRST.

From photos only: parts and placement. Assumed: I2C level shifter (Q2/Q3), buzzer driver details and supply rail, button pull−up, SHTC3 rail, capacitor−to−pin mapping, sensor pin numbers.

Unknown: remaining passive values, crystal pins and frequency, pad function of the 8−pin IC near the USB connector (not drawn), whether the dark−red level uses a blinking top LED.

Nets between groups are connected by name (labels): NRST, SWIM, SDA_3V3, SCL_3V3, BUTTON_PE3, BUZZER_PE7, LED_*.

Reconstructed from photos; values TBD; MCU pins are functional placeholders

Sheet: /
File: schematic.kicad_sch

Title: Eurotronic Air Quality Sensor 700209 − reconstruction

Size: A3	Date: 2026−11−03	**Rev: draft 0.4**
KiCad E.D.A. 9.0.8		Id: 1/1